COGNITIVE·TERRAIN

PRACTITIONER'S MANUAL · RUN IT · BUILD IT · KNOW WHY



RUN IT WITH A GROUP

Session plans from a lunch break to a multi-organisation exercise, and how to lead the debrief.

BUILD YOUR OWN CAMPAIGN

The campaign and technology editors, balance checks and sharing.

WHY IT WORKS THIS WAY

Design notes: what each mechanic models, and how it was arrived at.

EDUCATIONAL WARGAME · TURN-BASED
PLAYERS NEED ONLY THE FIELD MANUAL

CT · PM · 2026

FOR WHOEVER RUNS THE ROOM

BEFORE YOU BRIEF

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WHO THIS IS FOR

Instructors, exercise designers and facilitators who run Cognitive Terrain for other people, build campaigns for their own audience, or need to explain why the game models cognitive warfare the way it does.

Players need only the **Field Manual**, which is also built into the game as the Guide (**?**). This manual assumes you have read it or played a campaign.

WHAT PLAYERS SHOULD LEAVE WITH

The target of cognitive warfare is the quality of the other side's decisions, not territory or attention.

Reach is not effect: a viral post on the wrong area changes nothing that matters.

Attack and defence happen at once, and open and closed societies are vulnerable in different ways.

FIRST TIME RUNNING IT?

Play **Typhoon Ines** on **Standard** once yourself, end to end, before you run it for anyone else. It takes about forty minutes.

FOUR FORMATS · SUGGESTED TIMINGS

RUNNING A SESSION

The game runs offline on any computer with a browser, so the format is set by your time and your room, not by the technology. Timings are suggestions; adjust them to your group.

SOLO

ONE PLAYER · 30–45 MIN

1. Send the link or the HTML file with the Field Manual.
2. Ask for **Typhoon Ines** on **Recruit** with the tutorial first.
3. Ask them to bring a screenshot of the debrief and one turning point from the replay.

CLASSROOM

GROUPS OF 2–3 · 50–90 MIN

1. 10 min: the big idea (Field Manual page 3) on the projector.
2. 30–60 min: each group plays one campaign on one screen, one driver, others advising.
3. 15–20 min: debrief as a room (page 5 of this manual).

PAIRED SEMINAR

BLUE vs RED · 60–90 MIN

1. Pair students; one plays Blue, one Red, on two computers in the same room.
2. They swap turn codes by chat (page 6). Sixteen weeks takes about an hour.
3. Debrief each pair, then compare how Red was played across pairs.

DISTRIBUTED EXERCISE

SEVERAL ORGANISATIONS · DAYS

1. Build or choose a campaign and send the file to every cell first (page 9). Add a White Cell to steer it (page 7).
2. Cells play one week per day or per session, codes by email.
3. Collect debriefs and replays for a joint hot-wash.

WHAT TO DECIDE BEFOREHAND

Plain or military terms. Plain words suit mixed audiences; military terms (OODA, DQ, Seed Channels) suit staff courses. Players switch in the top bar.

Campaign and difficulty. See page 4.

Week in review on or off. On for learning; off for a faster, harder exercise.

KEEP IT MOVING

Set Red's turn speed to **FAST** once players know what Red's moves look like, and result pop-ups to **KEY ONLY**. Both are in Settings and change nothing in the rules.

EIGHT CAMPAIGNS · THREE DIFFICULTIES

CHOOSING A CAMPAIGN

Every campaign uses the same rules. What changes is the setting, the opponent, the twist at the start and which areas the four crises test. Pick by what you want the room to argue about.

CAMPAIGN	VS	TWIST AT THE START	CRISES TEST	GOOD FOR
Typhoon Ines	Olvana	None	Notice · Interpret · Act · Trust	First games; the core loop
The Valmora Vote	Olvana	Both TRUST areas weakened	Interpret · Notice · Decide · Trust	Elections, polarisation
Collision at Varrin Reef	Olvana	Both officials' entry points weakened	Notice · Act · Act · Decide	Crisis diplomacy, competing videos
Fever in Port Sabre	Olvana	Rumours spread faster for both; both NOTICE areas strained	Interpret · Trust · Act · Decide	Infodemics, public health
Winter Grid	Donovia	Blue's FOLLOW-THROUGH strained; 4 dormant narratives	Notice · Decide · Act · Trust	Energy coercion
The Lake Referendum	Donovia	Donovia already in Blue's INTERPRET; TRUST weakened	Interpret · Notice · Decide · Trust	Referendums, planted premises
The Silent Survey Ship	Donovia	Both DECIDE areas strained	Notice · Act · Interpret · Decide	Ambiguity, alliance choices
The Vell Valley	Donovia	5 dormant narratives; Blue's NOTICE and TRUST strained	Interpret · Act · Trust · Decide	Annexation and its cover story

DIFFICULTY

LEVEL	WHAT CHANGES
Recruit	Red is slower, makes more mistakes and earns 2 fewer points a week. For first games and short sessions.
Standard	A steady, sensible Red. Campaigns are balanced at this level.
Veteran	Red plans two moves ahead. No extra budget, no hidden information. For experienced players and competitions.

MATCH THE OPPONENT TO THE LESSON

Olvana teaches patience against nerve: a closed state that works one area at a time from inside.

Donovia teaches that the aim can be doubt and inaction rather than persuasion: floods, planted premises, shows of force.

Field Manual pages 24 and 25 set out each one's rules and sources.

SAME MAP, FAIR COMPARISON

Groups playing the same campaign get different maps unless they start from the same seed. For a like-for-like comparison, run it as a two-player game from one Blue code (page 6).

THREE VERDICTS · THE REPLAY · QUESTIONS

LEADING THE DEBRIEF

The game ends with a neutral debrief and a replay of both minds week by week. Most of the learning happens here, so leave time for it.

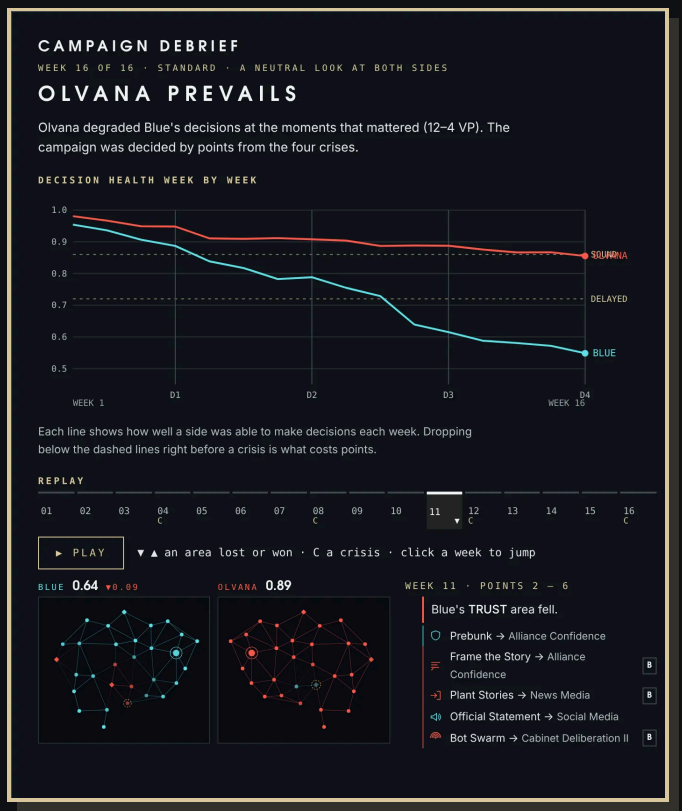


FIG. 1 The debrief: crisis points and health week by week, followed by three verdicts, turning points and lessons further down.

THE THREE VERDICTS

VERDICT	MEASURES
Who decided better	Average Decision Health across the campaign, week by week.
Who adapted more	How many of the eight capabilities each side used, and how many adaptive or protective moves.
Who got more for less	Influence moved per point spent attacking.

A side can win on points and lose two of the three verdicts. That gap is often the best conversation in the room.



FIG. 2 The replay: step through every week, with both minds and that week's moves.

QUESTIONS THAT WORK

- Which crisis did you lose, and when did you first know it was coming?
- Find a move with a great spread score that changed nothing. Why?
- When did defending your own mind matter more than attacking theirs?
- What did Red seem to be trying to do? What told you?
- Which of Blue's advantages as an open society did you use, and which did you lose?
- What would you tell a decision-maker in the real version of this crisis?

TWO PLAYERS · NO SERVER · NO SHARED NETWORK

ACROSS ORGANISATIONS

Two-player games need nothing but a way to send a line of text. That makes the game usable between cells, units or agencies that cannot share a network or install software.

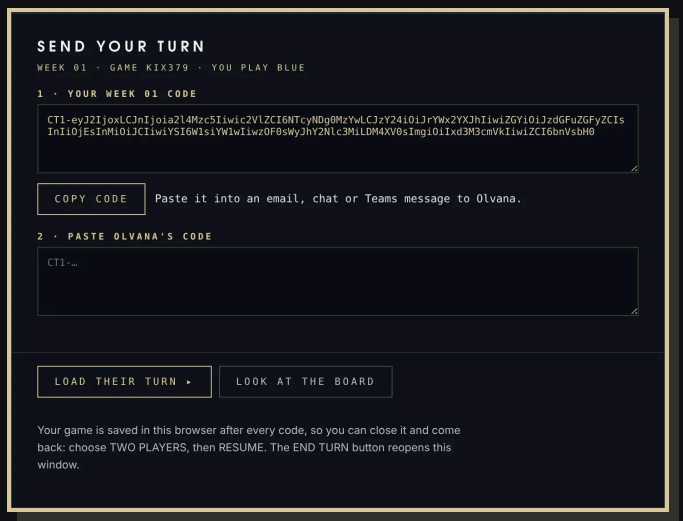


FIG. 3 A turn code with a COPY button, and a box for the reply.

WHAT A CODE CARRIES

The game and its map seed, the campaign, the week, the side and that side's moves. Each copy replays the other's moves and checks the board matches; if it ever does not, the log says so. Codes contain no personal data.

CHECKLIST

1. **Same version.** Codes work only between builds with the same minor version (0.17.x with 0.17.x), and the game cannot check this for you: a mismatch shows only as a failed board check. Send everyone the same download.
2. **Same campaign.** For a custom campaign, every Red player imports the file before the first code (page 9). The code checks it.
3. **Assign sides.** Blue starts the game; Red joins with Blue's first code. The campaign decides whether Red is Olvana or Donovia. To steer it yourself, start a structured session and join as the White Cell (page 7).
4. **Set a rhythm.** One week per session or per day keeps a sixteen-week campaign moving.
5. **Collect the debriefs.** Each side sees the same debrief at the end; Red sends one final code so Blue's game ends too (in a structured session, the White Cell sends it).

ANY CHANNEL WORKS

Email, chat, Teams, a text file on removable media. Extra text around a pasted code is ignored, so players can add a note.

Each player's game saves after every code and appears under **RESUME**, so a game can pause for days.

STRUCTURED SESSIONS · A THIRD PLAYER WHO STEERS

THE WHITE CELL

*In a structured session a facilitator joins as the **White Cell**. It sees both minds, and once a week, after Red and before Blue, it can adjust the game: run a story, change a budget, move a crisis, strike a move. It is the game's version of the control cell in a staff exercise.*

BLUE Plays the week → Code to Red → **RED** Plays the week → Code to White Cell → **WHITE CELL** Watches both, then steers → Code to Blue → **BLUE** Reads the changes, plays → Next week begins

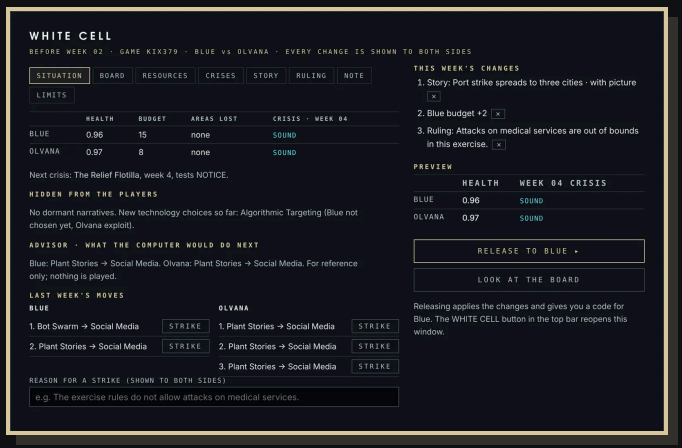


FIG. 4 The White Cell console before week 2: both sides' health, budget and next crisis; what is hidden from the players; last week's moves with STRIKE buttons; this week's changes and their effect on the right.

ONLY WHEN YOU CHOOSE IT

Ordinary two-player games are unchanged. The White Cell exists only when Blue starts a **NEW STRUCTURED SESSION · WITH WHITE CELL**.

SETTING ONE UP

1. **Blue starts it.** TWO PLAYERS → NEW STRUCTURED SESSION. Blue plays week 1 and sends the code to Red.
2. **Red joins** with Blue's code, as in any two-player game, and sends its code to the White Cell, not to Blue.
3. **The White Cell joins** with Red's first code: TWO PLAYERS → JOIN AS WHITE CELL. Both week 1 turns replay and the console opens.
4. **Release.** Queue changes, or none, and press RELEASE TO BLUE. Send the code to Blue. Week 2 begins.

WHAT THE WHITE CELL SEES

Both minds with no fog, both budgets, every move as it replays, Red's secret technology choices and Blue's dormant narratives. It plays no moves of its own.

The **WHITE CELL** button (where END TURN usually is) reopens the console; **LOOK AT THE BOARD** closes it to study the map.

WHAT THE PLAYERS SEE

Nothing is secret. Every story, ruling and change appears in a White Cell notice (page 8): Blue sees it before its turn, Red when Blue's code arrives. It is also in the log, the debrief and the replay.

THE CONSOLE · ONCE A WEEK · EVERY CHANGE VISIBLE

STEERING THE EXERCISE

Eight tabs, one queue, almost all of it buttons. Each change is listed on the right as you add it, with a preview of both sides' health and the next crisis. Remove any item before you release; nothing happens until you do.

SITUATION
Health, budgets, the next crisis, hidden facts, and what the computer would play next. STRIKE a move here.

BOARD
Pick a node on the map, then one click to shift control, set or clear a status, or plant a dormant narrative.

RESOURCES
One click to add or take budget points, or give either side a move more or fewer this week.

CRISES
Move, retarget or double a coming crisis, add a new one, or schedule a new technology.

STORY
A breaking news item, with an optional picture. Blue reads it before its turn, Red with Blue's next code.

RULING
An adjudication the players must follow, with your reason.

NOTE
A teaching note kept out of play; it appears in the debrief and replay.

LIMITS
Your own caps: largest shift and budget change; switch moves, crises or strikes off.



FIG. 5 What both players see before Blue's week 2: the story and its picture, the ruling and the changes, in plain words.

RULES THE GAME KEEPS FOR YOU

Control stays between −100 and 100; budgets between 0 and the cap; statuses last at most four weeks. Crises cannot move into the past or onto another crisis. Dormant narratives live only in Blue's mind.

A **strike** rewrites a week already played: every copy rebuilds itself without that move, and both sides are told why.

INJECTS THAT TEACH

TO SHOW	TRY
Events outrun preparation	Bring the next crisis forward a week, or make it test the area a side has neglected.
An outside shock	A story plus a small control shift on the matching node, e.g. a viral clip and a nudge on Social Media.
Resource pressure	Cut a side's budget by 2 before a crisis and watch what it gives up.
Rules of engagement	A ruling, then strike any move that breaks it.
A losing side recovers	One extra move for a week; ask in the debrief what it did with it.

POINT AND CLICK

Click any node on the map and the console opens on it, ready for a shift or a status. Drop, paste or browse for a picture to run with a story: it is shrunk to fit inside the code, so it travels by email or chat like any other code.

CAMPAIGN EDITOR · SHARE · IMPORT

MAKE A CAMPAIGN

Set the scene for your own audience, place the crises and new technologies on the calendar, tune how strong each side starts, then test it, check its balance and share it. No code involved.

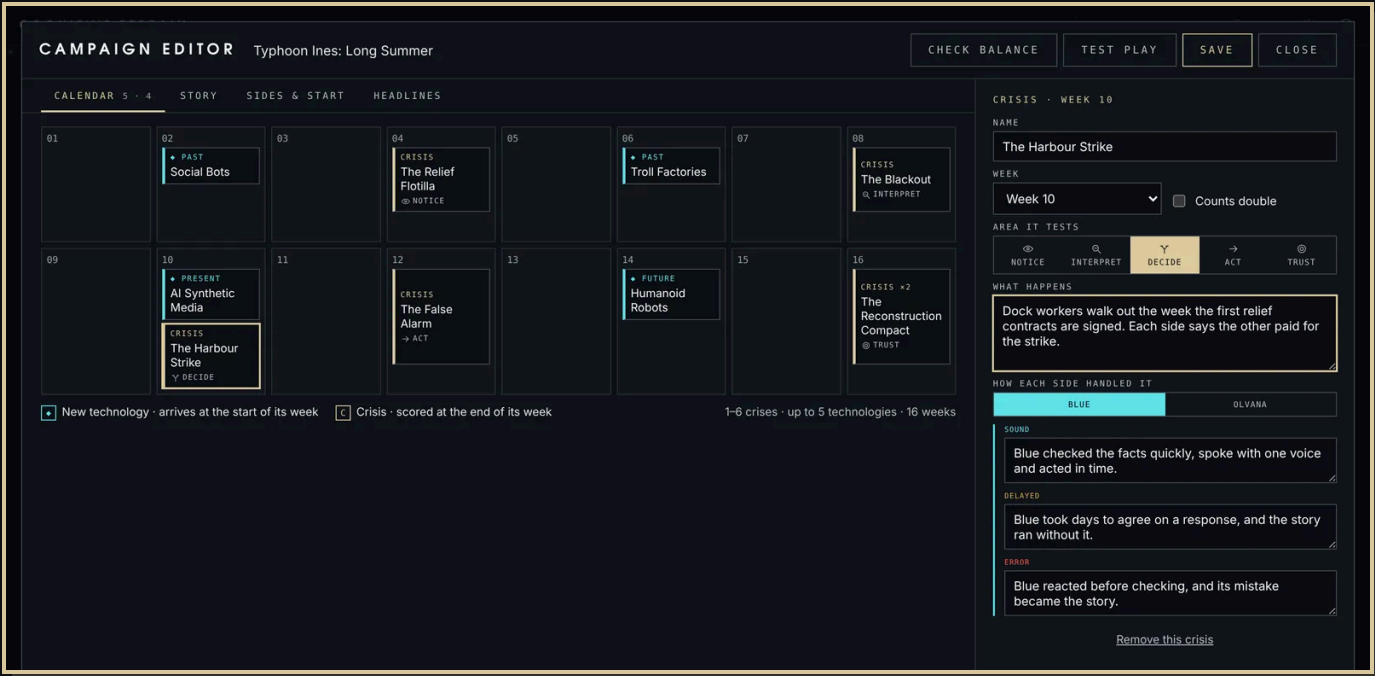


FIG. 6 The campaign editor, editing a copy of Typhoon Ines with a new crisis in week 10. Click a crisis or technology on the calendar to edit it on the right; + adds one in that week.

MAKE ONE

1. On the start screen press **CAMPAIGNS** (or ⚙ → **CAMPAIGN EDITOR**).
2. **+ NEW CAMPAIGN**, or **COPY & EDIT** a built-in one. Copying is the fastest way to a balanced start.
3. Choose the opponent (Olvana or Donovia), write the setting and the headlines' places and outlets, and set any twist at the start.
4. Place 1 to 6 crises and up to 5 new technologies (page 10).
5. **CHECK BALANCE**, then **TEST PLAY**.

SHARING

Save a **.campaign.json** file or copy a code that starts `CTC1-`. Campaigns are plain text and cannot run code, so they are safe to pass around. Campaigns live in the browser that made them: keep the file.



FIG. 7 The campaign library: play a campaign, or copy one to edit.

CRISES & TECHNOLOGIES

WRITING A CRISIS

A crisis is a moment that tests one area of both minds. For each one you write what happens, choose the area it tests, and write how each side handles it at each result: SOUND, DELAYED and ERROR. Players read your text at the moment of truth, so keep each line to one or two sentences about a decision, not an outcome.

> **Pick the area by the decision.** A flood of rumours tests NOTICE; a contested video tests INTERPRET; a deadline tests DECIDE; a delivery that must happen tests ACT; a scandal tests TRUST.

In the built-in campaigns the last crisis counts double; in your own, tick **Counts double** on any crisis. Spread the tested areas so no single defence wins the campaign.

CHECKING BALANCE

CHECK BALANCE plays your campaign 40 times, computer against computer, and reports how often each side wins. The built-in campaigns are tuned so Blue wins roughly 40–60% of games. If yours is far outside that, soften the twist or move a crisis to an area the stronger side has not prepared.

ORIGINAL FICTION

The built-in campaigns invent every place, outlet, person and event. Doing the same keeps a session about methods rather than about a real event, and keeps it usable in any room.

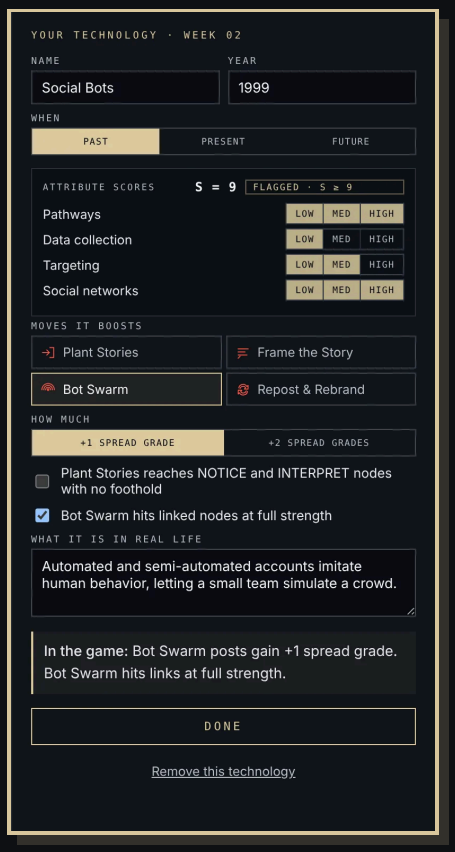


FIG. 8 Building a technology: name, year, era, the four attribute scores and their total S, and which moves it boosts.

TECHNOLOGIES

Pick from 18 real technologies, past, present and future, or **BUILD YOUR OWN**. Each is scored 1 to 3 on four attributes from Rushing, Umeokolo & Xu (2025): new pathways, data collection, targeting and social networks. A total of 9 or more is the paper's mark of a disruptive innovation. Where the paper gives no date or score, the editor marks the figure as the designer's estimate.

OFFLINE · PROJECTORS · SOUND · PRIVACY

SETTING UP THE ROOM

GETTING IT ONTO MACHINES

OPTION	BEST WHEN
Play in the browser (playtest page)	Internet is available and nothing can be installed.
One HTML file	Locked-down or offline machines: copy it over and double-click.
Installers (Windows, Mac, Linux)	Personal machines. Not yet code-signed: Windows needs More info → Run anyway; Mac needs right-click → Open the first time.

ON A PROJECTOR

Settings → **Chart Room** (the light scheme) reads best on washed-out projectors. Set **Interface size** to LARGER or LARGEST for the back row, and turn on **Colour-safe mode** so the sides differ by shape as well as colour.

PRIVACY

The game has no accounts, no analytics and no server; its only network request is for web fonts, and it works without them. Settings, saved two-player games, custom campaigns and a loaded voice pack stay in the browser on that machine. Clearing the browser's site data removes them.

IMMERSIVE BRIEFINGS IN A ROOM

The optional voice pack reads each week in review aloud as the broadcaster of the side ahead (Field Manual page 20). In a room it works best on the presenter's machine with speakers.

1. Download **CognitiveTerrain-Voice-Emma.zip** (about 111 MB) once, before the session.
2. Web or HTML file: Settings → Immersive Briefings → **LOAD VOICE PACK...**. Installers: put the zip beside the game before starting it.
3. Press **HEAR A SAMPLE** to check the speakers.

It runs offline, on that machine only. The voice is synthetic speech from an open-source model; tell your audience so. Slower machines take a few seconds before the first line.

ACCESSIBILITY CHECKLIST

Text size and interface size are separate settings. Every colour scheme passes 4.5:1 contrast. Reduced motion stops pulsing and drifting. Number keys pick moves; Esc cancels. Captions show whenever the narrator speaks.

DESIGN NOTES · KEVIN WILLIAMSON · 1 OF 4

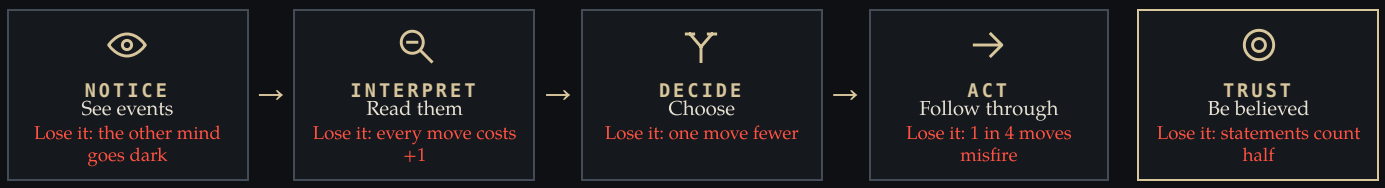
WHY IT WORKS THIS WAY

Each mechanic below says what it models, how I arrived at it, and where it comes from. Two kinds of tag show the difference at a glance.

FROM RESEARCH A published source shaped this rule.

DESIGNER'S ASSUMPTION My own design choice, made so the game plays well.

THE BOARD: FIVE AREAS OF ONE MIND



THE MIND AS TERRAIN

MODELS A nation's decision-making as the ground being fought over.

Two brains of about thirty connected nodes. Every node is a real kind of person or institution: a newsroom, an analysts' cell, a cabinet, a crisis team, the public's trust in its own government.

> **HOW I GOT THERE** *Cognitive warfare is hard to teach because you can't see it, and people picture it as "fake news". I wanted the real target on the table: whether the other side can still decide well under pressure.*

RUSHING, HERSCH & XU (2026)

LOSING AN AREA

MODELS What it costs to lose a capability.

An area falls when the other side holds half its nodes. Each loss has one plain effect (the strip above), so players feel it the way a staff would.

> **HOW I GOT THERE** *The five areas follow the research's cognitive OODA loop with trust underneath. The effects are deliberately simple so they can be read off the map.*

OODA FRAMING: RUSHING ET AL.

EFFECTS: DESIGNER'S ASSUMPTION

DECISION HEALTH

MODELS How well a side can still decide, 0 to 1.

The score is not views or nodes held. It is the health of the four loop areas, multiplied by trust, minus a penalty for each area lost. The math is on page 16.

> **HOW I GOT THERE** *Spreading far is not the same as changing minds. A centralized state leans 40% on DECIDE (Blue 30%): when everything runs through the top, the top is the target.*

RUSHING ET AL. · WEIGHTS AFTER SHIN (NIU)

WEIGHTS: DESIGNER'S READING

ENTRY POINTS AND FOOTHOLDS

MODELS Messages need a way in before they can travel.

Each brain has three entry points: News Media, Social Media and Government Contacts. Attacks start there and spread along links from nodes you have a foothold in.

> **HOW I GOT THERE** *It makes players plan an approach instead of hitting the most valuable node first, which is how influence work actually has to be sequenced.*

DESIGNER'S ASSUMPTION

> **"SPREADING FAR IS NOT THE SAME AS CHANGING MINDS."**

DESIGN NOTES · 2 OF 4 · MOVES, SPREAD AND FAILURE

MOVES & METRICS

Eight moves, one engagement metric, and odds that change when a side prepares.

Plant Stories | Frame the Story | Bot Swarm | Repost & Rebrand | Watch Team | Prebunk | Official Statement | Lessons Learned

EIGHT MOVES, EIGHT CAPABILITIES

MODELS The attacker's and defender's capability families.

Four attack cards and four defence cards, one for each capability family in the research. Military mode keeps doctrinal names. Donovia adds a ninth, Show of Force.

> **HOW I GOT THERE** *I did not invent a list of tricks. Mapping one card to each family keeps the game honest to the framework and easy to cite in a debrief.*

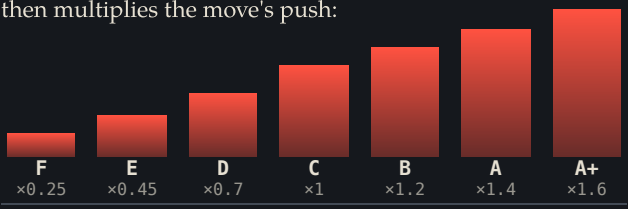
RUSHING, HERSCH & XU (2026) §2.2.2, §2.3.2

SHOW OF FORCE: ISW, IISS

THE SPREAD SCORE

MODELS How engaging a campaign of posts actually was.

Every attack except Show of Force rolls an audience and is graded with the research's engagement metric. The grade then multiplies the move's push:



> **HOW I GOT THERE** *I kept the formula and grade scale because they teach that interaction outweighs views. The multipliers are mine. A viral post on the wrong area still loses the crisis.*

METRIC AND GRADES: RUSHING & XU (2025)

MULTIPLIERS: DESIGNER'S ASSUMPTION

FAILURE AND BACKFIRE

MODELS Platforms, watchers and prepared audiences.

Any attack can be taken down or fall flat. Watching and prebunking raise those odds, and a failed attack nudges the node back toward the defender (page 17 has the numbers).

> **HOW I GOT THERE** *I wanted defence to change the odds, not just add hit points, so players feel why detection and inoculation matter before a campaign lands.*

DESIGNER'S ASSUMPTION

RECOVERY, FRAMING AND SPREAD

MODELS Minds drifting back, and ideas settling in.

Each week every node recovers a little toward its own side; strongly held nodes push on their neighbours. Framing stops recovery for two weeks (three when Donovia frames), so the next push sticks: the game's version of priming.

> **HOW I GOT THERE** *Tuned so that a single hit fades but a sustained, sequenced campaign holds.*

DESIGNER'S ASSUMPTION

> **"DEFENCE SHOULD CHANGE THE ODDS, NOT JUST ADD HIT POINTS."**

DESIGN NOTES · 3 OF 4 · TIME, CRISES AND TECHNOLOGY

THE MOMENTS THAT COUNT

Sixteen weeks so it fits a lunch break; four crises so preparation pays.

01	02 ◆ TECH	03	04 CRISIS	05	06 ◆ TECH	07	08 CRISIS	09	10 ◆ TECH	11	12 CRISIS	13	14 ◆ TECH	15	16 CRISIS ×2

⚠ FOUR CRISES

MODELS Decisions under pressure at the moments that matter.

In weeks 4, 8, 12 and 16 a crisis tests one area of both minds; in the built-in campaigns the last counts double. The result depends on overall health and, heavily, on the tested area (page 16).

> **HOW I GOT THERE** *Weighting the tested area makes players read the calendar and prepare, which is the skill I wanted to reward.*

OODA FRAMING: RUSHING ET AL.

SCORING: DESIGNER'S ASSUMPTION

COGNITIVE SUPERIORITY

MODELS One side's decision-making clearly dominating.

If one side has lost at least two more areas than the other for two weeks running, the campaign ends early. The race shows in the side panel as LEAD.

> **HOW I GOT THERE** *The research defines cognitive superiority as a sustained advantage in the decision loop; the knockout gives that idea a hard edge.*

RUSHING, HERSCH & XU (2026)

THRESHOLDS: DESIGNER'S ASSUMPTION

NEW TECHNOLOGIES

MODELS Disruptive innovations arriving mid-campaign.

Both sides choose in secret: use it for an attack bonus, or prepare and deny the other side's. Using it against a prepared opponent gains nothing.

> **HOW I GOT THERE** *A dilemma, because that is the real problem with new tools: you don't know whether the other side is ready. The scoring and the threshold of 9 are the paper's.*

RUSHING, UMEOKOLO & XU (2025)

SOME SCORES: DESIGNER'S ESTIMATE

BUDGET AND MOVES

MODELS Finite attention and resources.

Three moves a week from a budget that grows with what you hold, capped at 18, with one extra move for points left over. Access to officials pays more than any post.

> **HOW I GOT THERE** *The cap stops hoarding; the extra move rewards spending.*

DESIGNER'S ASSUMPTION

> **"READ THE CALENDAR, AND PREPARE THE AREA THE NEXT CRISIS TESTS."**

RED IS NEVER A MIRROR

	BLUE	OLVANA	DONOVIA
IS	An open democracy	A closed, one-party state	An oligarchic state at permanent information war
STRENGTH	Free press, truth that sticks, cheap extra move	Censored internet, state-funded content, already inside	Floods, planted premises, dormant narratives, shows of force
WEAKNESS	Its open internet lets Red's posts spread	Runs from the top; slow to learn	Predictable; threats can blow back
SHOULD FEEL	Under pressure, keeping its head	A fortress with one gate	Loud

OPEN AGAINST CLOSED

MODELS Why open and closed societies are vulnerable in different ways.

Blue's posts into a censored internet spread less; Red's into Blue's open internet spread further. Blue gets trust that recovers, statements that hit harder while trust holds, and truth that sticks.

> **HOW I GOT THERE** *The asymmetry is the point: you fight a closed system differently from how it fights you.*

ATP 3-13.1 · SHIN (NIU)

SOME EFFECTS: DESIGNER'S ASSUMPTION

TWO OPPONENTS, KEPT FICTIONAL

MODELS Two doctrines of cognitive warfare.

Red's countries come from the Army's training world, so they are familiar and clearly fictional. Every rule names its source or is labelled as my assumption; every claim was checked against the sources in 0.14.2.

> **HOW I GOT THERE** *Both share a core and differ in temperament, so each feels different with a person playing it.*

ODIN DATE · NIU · EADS ET AL.

NATO C2COE · ISW · IISS · CARNEGIE

AN OPPONENT THAT PLAYS FAIR

MODELS A competent adversary at three levels of skill.

Recruit is noisier and earns 2 fewer points a week. Veteran plans two moves ahead on a scratch copy of the game: no extra budget, no hidden information, never the game's own dice.

> **HOW I GOT THERE** *Very hard to beat, without cheating. Each of its moves also shows the likely reason, judged only from what the player could see.*

DESIGNER'S ASSUMPTION

BUILT TO BE USED

MODELS A teaching game that runs anywhere.

It explains itself at every moment that matters. Two players need only a turn code; campaigns are plain text that cannot run code; Immersive Briefings are optional; every scheme passes 4.5:1 contrast.

> **HOW I GOT THERE** *Keep the original game: new features go around the core loop, never instead of it. A campaign is balanced when Blue wins 40–60% of computer games.*

DESIGNER'S ASSUMPTION

> **"VERY HARD TO BEAT, WITHOUT CHEATING."**

THE MATH, SHOWN

Every number the game uses to score a campaign, with a worked example. Nothing here is hidden from players; the side panel and the week in review show the same sums.

1 · AREA HEALTH

```
area = average over its nodes of
clamp( (lean + 50) ÷ 150, 0, 1 )
```

A node's lean runs from −100 (held hard by the enemy) to +100 (held hard by its owner). A node fully yours counts 1; a neutral one about 0.33; one the enemy holds at −50 or worse counts 0. Partial credit keeps the score from lurching.

2 · DECISION HEALTH

```
areas = w·NOTICE + w·INTERPRET + w·DECIDE + w·ACT
DQ = areas × ( 0.75 + 0.25 · TRUST )
     − 0.06 × areas lost − flood
```

WEIGHT	NOTICE	INTERPRET	DECIDE	ACT
BLUE	0.20	0.25	0.30	0.25
OLVANA · DONOVIA	0.15	0.20	0.40	0.25

Trust scales everything: with trust gone, the most a side can score is 0.75. **Flood** (Donovia only): −0.02 for each of its attacks that lands in a week beyond the first two (one more is free per Lessons Learned level), at most −0.06, until Donovia's next turn.

WHAT MOVES THE NEEDLE

CHANGE	DECISION HEALTH	NEXT CRISIS
Tested area +0.10	+0.01 to +0.04	**+0.05 to +0.06**
Any other area +0.10	+0.01 to +0.04	+0.01 to +0.02
TRUST +0.10	about +0.02	about +0.01 (more if the crisis tests TRUST)
Lose an area	−0.06 and more	−0.04; −0.12 if tested

The lesson in numbers: the area the next crisis tests is worth about three times any other. That is why Looking ahead names it.

3 · A CRISIS

```
crisis = 0.6 · DQ + 0.4 · tested area
− 0.08 if the tested area is lost
```

ERROR	DELAYED	SOUND
below 0.72 · 0 pts	0.72+ · 1 pt	0.86+ · 3 pts

Plus 1 to a side whenever the other side scores ERROR. A crisis that counts double (week 16 in the built-in campaigns) doubles everything, bonus included.

The "decides in ... hours" shown in the game is 6 ÷ score², rounded.

WORKED EXAMPLE

Blue's areas: NOTICE 0.9, INTERPRET 0.8, DECIDE 1.0, ACT 0.9; TRUST 0.7; nothing lost.

```
areas = .2×.9 + .25×.8 + .3×1 +
.25×.9 = 0.905
trust = 0.75 + 0.25×0.7 = 0.925
DQ = 0.905 × 0.925 = 0.84
```

A crisis testing INTERPRET: 0.6×0.84 + 0.4×0.8 = **0.82**, so **DELAYED**, 1 point. Lift INTERPRET to 0.9 and DQ rises too: about **0.88**, **SOUND**, 3 points.

4 · WINNING

Most crisis points after week 16 wins. **Tied?** The side whose Decision Health is more than 0.03 higher wins; otherwise it is a draw. **Knockout** earlier: the other side has lost at least 2 areas, and at least 2 more than you, at the end of 2 weeks running.

ODDS & BALANCE

How hard a move hits, how often it fails, what comes in each week, and how far to trust a balance check.

5 · SPREAD SCORE AND GRADE

```
score = ( 0.1·views + 0.3·likes +
0.7·comments + 1·shares ) ÷ posts
```

GRADE	SCORE	PUSH ×
F · E · D	0–2 · 3 · 4–9	0.25 · 0.45 · 0.7
C · B	10+ · 100+	1.0 · 1.2
A · A+	1,000+ · 10,000+	1.4 · 1.6

Reach is rolled on a ten-times scale, so ×10 is about one grade. **About a grade up:** their TRUST lost; each step of a technology you used. **About a grade down:** a Watch Team on the target. **Smaller nudges:** an entry point ×2.5; holding their News Media ×3; Red posting into Blue's open internet ×1.25; Blue posting into Olvana's censored internet ÷1.4 (Donovia's ÷1.25).

6 · PUSH

```
push = card power × area fit × grade
× 0.5 if prebunked (0.45 for Donovia's
attacks)
× 1.4 if framed × 2 if it wakes a dormant
narrative
```

Card power: Plant Stories 20 (×1.5 on an entry point), Frame 16, Bot Swarm 12 plus 7 on every link, Repost 10, Show of Force 16 (no audience roll, so no grade). **Area fit** runs 0.75 to 1.5: a card's best area (Field Manual Quick Reference) gets 1.5; Repost & Rebrand is 1 everywhere.

> **Example.** Plant Stories on their News Media (an entry point, in NOTICE, the card's best area), graded B: 20 × 1.5 × 1.5 × 1.2 = **54**. Lean runs −100 to +100: a node is **held** at 30 toward a side and gives a **foothold** at 10.

7 · ODDS OF FAILURE

ATTACK FAILS IF…	CHANCE
Any attack	8%
+ target watched	+15%
+ target prebunked	+8%
+ Blue posting into Olvana / Donovia	+2% / +1%
Show of Force	+4%
Official Statement ignored	6% (+20% if TRUST lost)

A failed attack pushes the node back 6 toward the defender (9 when Donovia's fail); a failed Show of Force also pushes each neighbour back 4. With ACT lost, 1 move in 4 misfires before any of this is rolled.

8 · EACH WEEK

Budget: 6 + 1 per enemy node held (Top Leadership 2) + 2 for their Government Contacts; income at least 2 a week; a side can hold at most 18 (the rest is wasted). Recruit Red −2; Donovia +1. Extra move: Blue 2, Red 5.

Recovery: each node +1 toward its owner (+1 per Lessons Learned level, half for Red), halved with TRUST lost. Blue's TRUST nodes +2 more; Red's entry points ×1.2 (Donovia's ×1.1); Red's nodes leaning Blue only ×0.35. None while framed or once captured. Any neighbour leaning 50+ the other way pushes 5 (half if prebunked).

9 · READING A BALANCE CHECK

The target is Blue winning 40–60% of computer games. **CHECK BALANCE** plays 40, so a result is only good to about ±15 points: 35% might really be 50%. Treat 30–70% as "probably fine", and run it twice before changing anything.

BLUE WON (OF 40)	LIKELY TRUE RANGE
12 · 30%	16–44%
20 · 50%	35–65%
28 · 70%	56–84%

95% ranges. For a firmer answer, run CHECK BALANCE several times and add up the wins.

WHERE THIS COMES FROM

The game is built on published research. The rules simplify it; the debrief and the in-game guide point back to it.

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ATTRIBUTION AND LICENSE

This game and manual adapt concepts, the capability taxonomy, the engagement-effectiveness metric (formula, interaction weights and grade scale) and two sets of innovation attribute scores from the three Rushing et al. papers above. Each paper is licensed under Creative Commons Attribution 4.0 International (CC BY 4.0): creativecommons.org/licenses/by/4.0. Papers: arxiv.org/abs/2603.05222 · arxiv.org/abs/2510.15805 · arxiv.org/abs/2510.15801.

Changes made: the material was adapted into game mechanics and rewritten for play. Engagement grades are mapped to effect multipliers invented for this game; the OODA effects, capability families and cognitive superiority definition are expressed as game rules; 16 of the 18 technology score sets are the designer's own estimates (the paper scores AI Synthetic Media and Humanoid Robots); all scenario, narrative and tutorial text is original.

No endorsement: the authors and their institutions have not reviewed or endorsed this game.

Immersive Briefings voice pack (optional, separate download): Kokoro-82M by hexgrad, kokoro-js and transformers.js (Apache 2.0); ONNX Runtime (MIT); eSpeak NG (GPL-3.0-or-later), whose source is published with the pack.

HOW THIS GAME WAS MADE

Design and direction: Kevin Williamson. Cognitive Terrain is my design: I conceived the game, built its rules from the research, chose every source, defined both opponents and the campaign briefs, set the balance targets and accessibility standards, and reviewed every rule, campaign and page.

Tools: I used AI-assisted development tools to help implement the code, draft text and run balance tests, working from my specifications; I reviewed, revised or rejected everything they produced. The published sources, not the tools, are the authority for any rule that models real-world behaviour. The art, icons and music are drawn and synthesized by the game's own code; no AI image or music generators were used, and the base game runs no AI while you play. The optional Immersive Briefings voice is synthetic speech from an open-source model that runs on your own computer (Field Manual page 20).

Olvana and Donovia, and the countries Bothnia, Framland, Arnland and Gorgas, are fictional training countries from the US Army's ODIN DATE environment; they are used here without endorsement. All eight scenarios, their towns, outlets, people and every event in them are original fiction. Any resemblance to real places or events is coincidental. Donovia's rules follow the NATO, ISW, IISS and Carnegie research above; game effects are abstractions of broad, publicly reported patterns, not claims about any specific operation.